

Minimize loss function of regression by fundamental lemma of calculus of variation

$$E[L] = \iint (y(x) - t)^2 p(x, t) dx dt$$

1. $y \rightarrow y + \epsilon \nu$ for any $\nu(x)$

$$\Rightarrow \iint (y + \epsilon \nu - t)^2 p(x, t) dx dt$$

2. partial derivate with respect to ϵ

$$\frac{\partial}{\partial \epsilon} \iint (y + \epsilon \nu - t)^2 p(x, t) dx dt = \iint 2(y + \epsilon \nu - t) \cdot \nu p(x, t) dx dt$$

$$3. p(x, t) = p(x) \cdot p(t|x)$$

$$\Rightarrow \iint 2(y + \epsilon \nu - t) \nu \cdot p(t|x) p(x) dx dt$$

4. $\epsilon \rightarrow 0$ and set integral to zero

$$\lim_{\epsilon \rightarrow 0} \iint 2(y + \epsilon \nu - t) \nu \cdot p(t|x) p(x) dx dt = 2 \iint (y - t) \nu p(t|x) p(x) dx dt$$

$$= 2 \int \underbrace{\nu(x) p(x)}_{\text{irrelevant with } t} \left[\int (y - t) p(t|x) dt \right] dx = 0$$

for any $\nu(x)$, see this integral to zero gives (FLCV) applying Fubini's theorem (how?)

$$\int (y - t) p(t|x) dt = 0 \Rightarrow \int y p(t|x) dt = \int t p(t|x) dt$$

$$\Rightarrow \int \underbrace{p(t|x)}_1 dt = E[t|x] \therefore \frac{\delta E[L]}{\delta y(x)} = 0 \Rightarrow y = E[t|x]$$